USE CASE STUDY REPORT

Group No.: Group 2

Student Names: Praniti Kale & Rutuja Dambir

Executive Summary:
The main goal of this project was to develop and implement a relational database to support the Boston Women Entrepreneurs ecosystem. Boston Women Entrepreneurs is a network of female entrepreneurs from MIT, Harvard, Northeastern University, Boston University and others who are building new technology companies, fashion brands, sustainable businesses, educational products and healthcare solutions. Boston area Incubators and Accelerators collect information about their entrepreneurs, start-ups, mentors, funding rounds, events and investors in many different formats, which can often result in duplicates or loss of important information. By creating a centralized, structured, and scalable relational database system (known as the Boston Women Entrepreneurs Incubator Tracker), Boston Women Entrepreneurs has been able to create a more efficient process to capture the full range of activity within the incubator network. The Incubator Tracker reduces the time required to retrieve information, eliminates duplication, and improves access to analytics and reports. Additionally, the system captures critical data points regarding women founders, startup progress, mentor quality, funding history, and event participation. As a result, the Incubator Tracker enables incubators to make data-driven decisions, and to actively support the success of female entrepreneurs.

The development of the database was based on an analysis of actual workflows of incubators and a review of the types of information collected during startup applications, mentor assignments, funding processes, and event registrations. As a result of this research, both an EER (Extended Entity Relationship) Diagram and UML (Unified Modeling Language) Model were created, as inputs to the next phase of development converting the high-level conceptual information to a relational database schema.

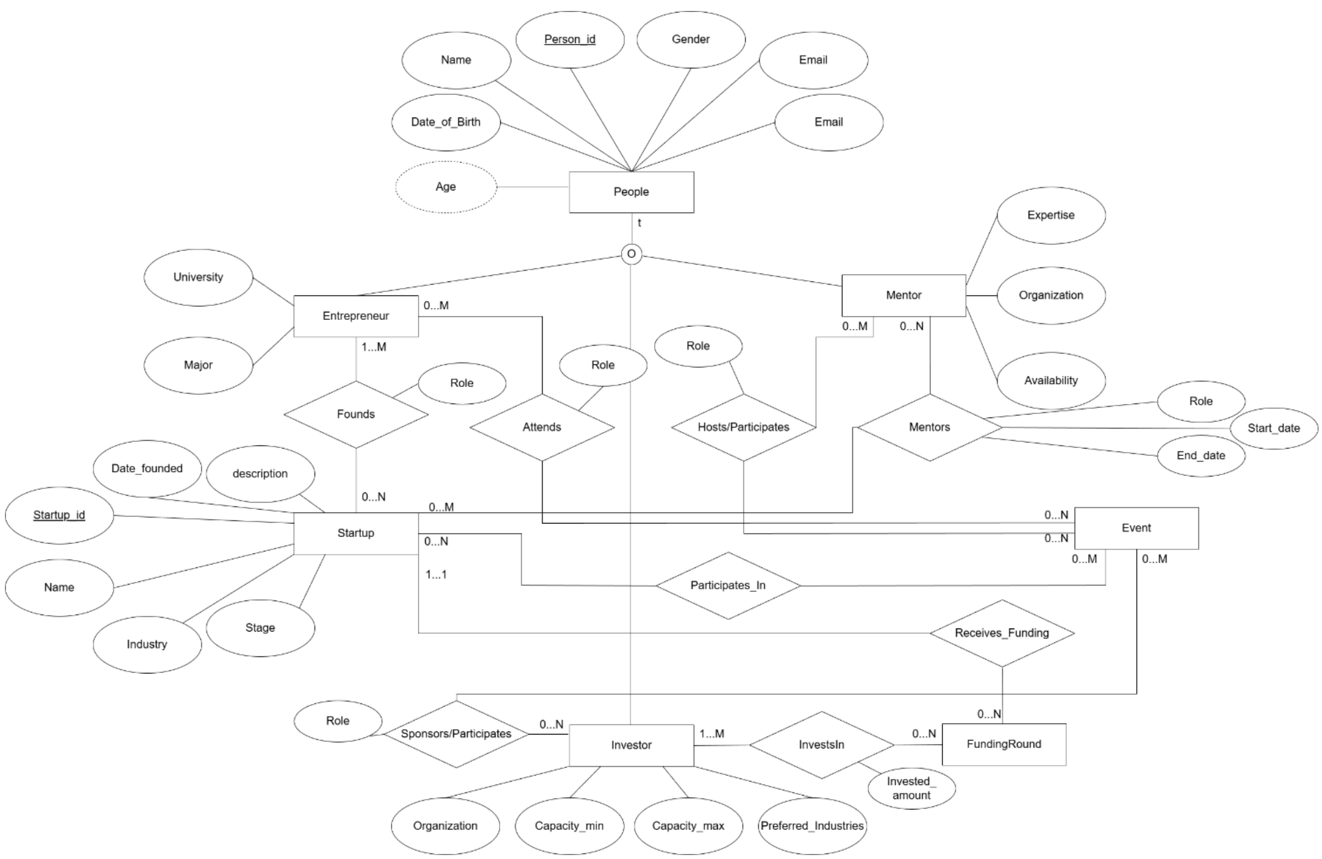
I. Introduction

Boston is the world's leading innovation center with over a thousand female students launching their entrepreneurship efforts across multiple industries, including tech, fashion, health, sustainability, and education. Female entrepreneurs in Boston receive support from many incube programs, accelerators and university programs through mentoring relationships, funding opportunities, and entrepreneurship programs. Unfortunately, there is currently no centralized database for incubator programs to aggregate activity data on women-led start-ups and incubators; instead, incubators have small, separate databases that do not allow for complete visibility, track and report aggregated programmes without fragmentation, create disparate data entry issues, and ultimately create an incomplete view of incubation activity within the incubators. As a result, female entrepreneurs in Boston have to manually compile multiple sources of data to manually generate analytics, which limits their ability to provide meaningful insights.

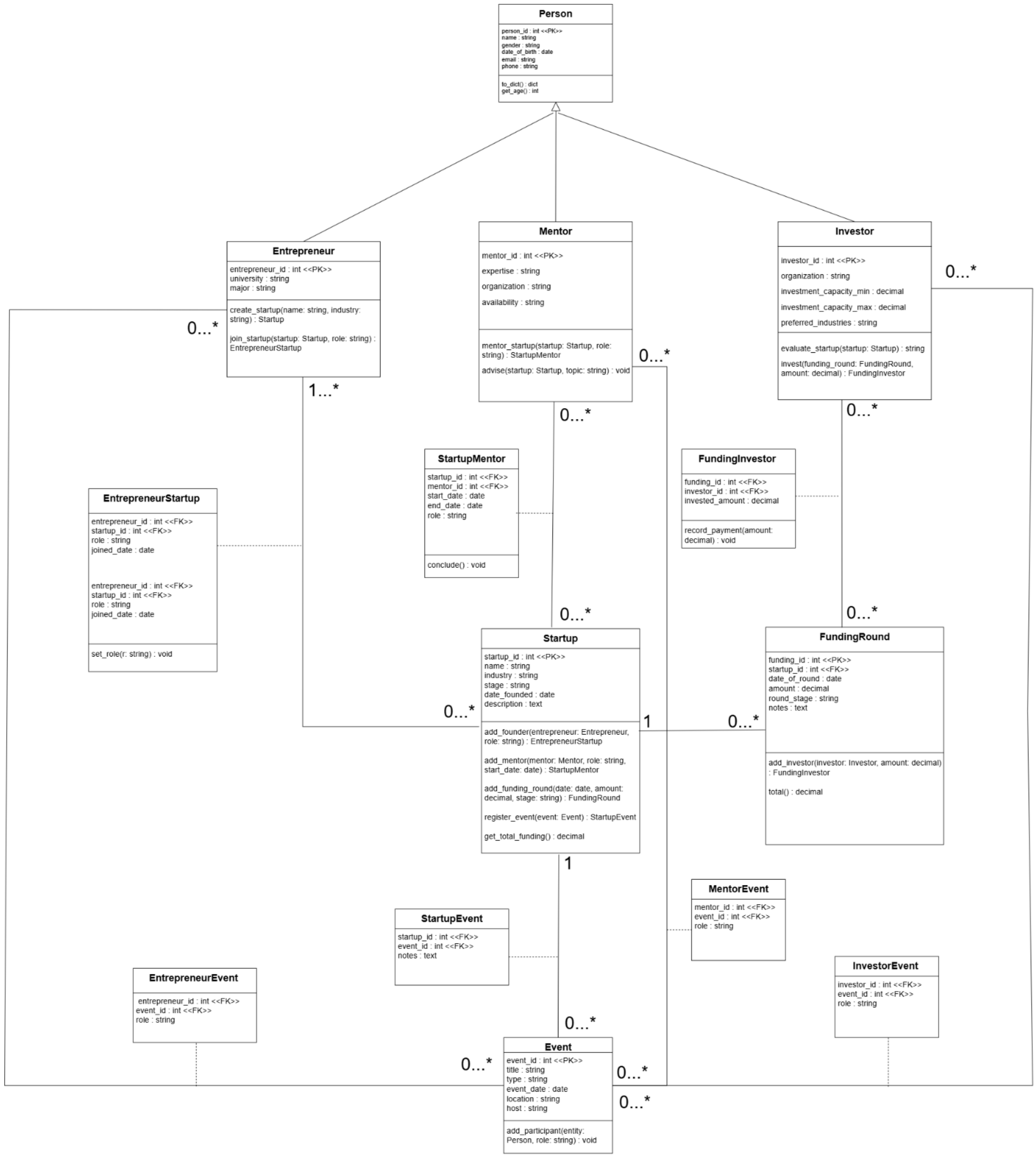
To address the unmet needs of female entrepreneurs by providing a single source of information that collects and aggregate all of the data about female entrepreneurs using a relational database. The mission of the Boston Women Entrepreneurship Incubator Tracker is to consolidate females' entrepreneurial activity data, including as many aspects of the start-up ecosystem (start-ups, entrepreneurs, mentors investors, events/funding and participant records) into one cohesive structure and standard format to create a more accurate and efficient incubator operation. All female entrepreneurs will have a unique ID and milestones for recording every start-up in the new system and tracking their progress through an automated milestone tracking tool.

II. Conceptual Data Modeling

1. EER Diagram



2. UML Diagram



III. Mapping Conceptual Model to Relational Model

Primary Key- **Underlined** **Foreign Key-** ***Italicized***

Superclass:

People(Person_ID, Name, Gender, Phone_No, Email, *Date_of_Birth*)

Subclass:

Person_ID: Foreign Key, refers to Person_ID in People, Not Null

Entrepreneur(Entrepreneur_ID,University, Major, *Person_ID*)

Mentor(Mentor_ID, Expertise, Organization, Availability, *Person_ID*)

Availability set as Available by Default

Investor(Investor_ID, Organization, capacity_min, capacity_max, Prefered_industry, *Person_ID*)

Event(Event_ID, Title, type, location, host, event_date)

Startup(Startup_ID, Name, Industry, Stage, Date_founded, description)

FundingRound(Funding_ID, date_of_round, amount, round_stage, notes, startup_ID)

Startup_ID: Foreign Key, refers to Startup_ID in Startup, Not Null

Founds(Entrepreneur_ID, Startup_id,role)

Entrepreneur_ID refers to Entrepreneur_ID in Entrepreneur

Startup_ID refers to Startup_ID in Startup

Entrepreneur_ID, Startup_ID primary key

Mentors_startup(Startup_ID, Mentor_ID, role, start_date, end_date)

Startup_ID refers to Startup_ID in Startup

Mentor_ID refers to Mentor_ID in Mentor

Mentor_ID, Startup_ID primary key

Invests_In(Funding_ID, Investor_ID, invested_amount,)

Funding_ID refers to Funding_ID in FundingRound

Investor_ID refers to Investor_ID in Investor

Funding_ID, Investor_ID primary key

Attends(<u>Enterpreneur_ID, Event_ID</u>, role)

Entrepreneur_ID refers to Entrepreneur_ID in Entrepreneur

Event_ID refers to Event_ID in Event

Entrepreneur_ID, Event_ID primary key

Participates(<u>Startup_ID, Event_ID</u>, role)

Startup_ID refers to Startup_ID in Startup

Event_ID refers to Event_ID in Event

Startup_ID, Event_ID primary key

Hosts/Participates(<u>Mentor_ID, Event_ID</u>, role)

Mentor_ID refers to Mentor_ID in Mentor

Event_ID refers to Event_ID in Event

Mentor_ID, Event_ID primary key

Sponsors/Participates(<u>Investor_ID, Event_ID</u>, role)

Investor_ID refers to Investor_ID in Investor

Event_ID refers to Event_ID in Event

Investor_ID, Event_ID primary key

IV. Implementation of Relation Model via MySQL and NoSQL

MySQL Implementation:

The database was created in MySQL and the following queries were performed:

1. AGGREGATE

Query: Full Funding Statistics Per Startup

```
SELECT
s.startup_id,
s.name AS startup_name,
SUM(fr.amount) AS total_funding,
COUNT(fr.funding_id) AS num_rounds,
AVG(fr.amount) AS avg_round_amount,
MIN(fr.amount) AS min_round_amount,
MAX(fr.amount) AS max_round_amount
FROM Startup s
LEFT JOIN FundingRound fr
ON fr.startup_id = s.startup_id
GROUP BY s.startup_id, s.name
ORDER BY total_funding DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

startup_id	startup_name	total_funding	num_rounds	avg_round_amount	min_round_amount	max_round_amount
47	SparkWorks47	6163067.00	5	1232613.400000	22623.00	2615912.00
70	BetaApps70	4640394.00	2	2320197.000000	1527811.00	3112583.00
73	AlphaLabs73	4567806.00	5	913561.200000	98557.00	2602052.00
93	EcoSystems93	4270085.00	3	1423361.666667	27036.00	2603499.00
74	AlphaWorks74	4080746.00	3	1360248.666667	8717.00	3387939.00
14	NextLabs14	3740752.00	6	623458.666667	8246.00	2511677.00
36	OmniSolutions36	3636773.00	3	1212257.666667	25773.00	3505109.00
38	OmniSolutions38	3582585.00	1	3582585.000000	3582585.00	3582585.00
58	AlphaLabs58	3422674.00	6	570445.666667	10818.00	2580893.00
102	BlueWorks102	3352218.00	3	1117406.000000	9099.00	3320532.00
99	NextLabs99	3312578.00	1	3312578.000000	3312578.00	3312578.00

2. INNER JOIN

Query: List mentors with their matched startups

```
SELECT p.name AS mentor_name, s.name AS startup_name, sm.role
FROM Mentor m
JOIN People p ON p.person_id = m.person_id
JOIN StartupMentor sm ON sm.mentor_id = m.mentor_id
JOIN Startup s ON s.startup_id = sm.startup_id
ORDER BY mentor_name;
```

mentor_name	startup_name	role
Abigail Carter	EcoLabs95	Advisor
Abigail Carter	NovaWorks91	Technical Mentor
Abigail Garcia	AlphaLabs33	Lead Mentor
Abigail Moore	BlueApps2	Lead Mentor
Abigail Moore	AlphaDesigns115	Advisor
Abigail Moore	BrightWorks90	Advisor
Abigail Scott	SparkDesigns94	Advisor
Abigail Scott	GreenApps7	Lead Mentor
Abigail Scott	SparkTech60	Lead Mentor
Abigail Scott	FusionSolutions83	Lead Mentor
Abigail Scott	EcoSystems93	Lead Mentor

3. NESTED QUERY (Subquery in WHERE clause)

Query: Entrepreneurs who founded more than one startup

```
SELECT e.entrepreneur_id, p.name
FROM Entrepreneur e
JOIN People p ON p.person_id = e.person_id
WHERE e.entrepreneur_id IN (
   SELECT entrepreneur_id
   FROM EntrepreneurStartup
   GROUP BY entrepreneur_id
   HAVING COUNT(startup_id) >
1
);
```

entrepreneur_id	name
9	Nisha Davis
35	Grace Wilson
27	Skylar Hill
64	Chloe Robinson
30	Isabella Phillips
76	Savannah Wilson
101	Lily Mitchell
93	Elena Johnson
97	Zoey Reyes
42	Bella Jones
79	Evelyn Lewis

4. CORRELATED SUBQUERY

Query: Events with above-average participation compared to their own event type

```
SELECT e.event_id, e.title, e.type,
       (SELECT COUNT(*)
        FROM EntrepreneurEvent ee
        WHERE ee.event_id = e.event_id) AS attendees
FROM Event e
WHERE (SELECT COUNT(*)
       FROM EntrepreneurEvent ee
       WHERE ee.event_id = e.event_id) >
      (SELECT AVG(x.attendee_count)
       FROM (
         SELECT event_id, COUNT(*) AS attendee_count
         FROM EntrepreneurEvent
         GROUP BY event_id
       ) x);
```

event_id	title	type	attendees
6	Harvard Workshop 6	Demo Day	5
8	Boston Pitch Day 8	Networking	11
9	Student Pitch Day 9	Demo Day	5
19	Student Investor Networking 19	Pitch	6
20	Student Investor Networking 20	Pitch	6
23	NU Demo Night 23	Networking	6
26	Student Investor Networking 26	Workshop	7
27	MIT Pitch Day 27	Workshop	7
30	NU Bootcamp 30	Investor Forum	6
31	Boston Demo Night 31	Pitch	6
32	MIT Workshop 32	Workshop	6

5. ALL

Query: Find startups whose total funding raised is greater than ALL investors' minimum investment capacity.

```
SELECT
    s.startup_id,
    s.name AS startup_name,
    SUM(fr.amount) AS total_funding
FROM Startup s
JOIN FundingRound fr ON fr.startup_id = s.startup_id
GROUP BY s.startup_id, s.name
HAVING SUM(fr.amount) > ALL (
       SELECT capacity_min
       FROM Investor
     );
```

startup_id	startup_name	total_funding
73	AlphaLabs73	4567806.00
7	GreenApps7	772242.00
100	BlueApps100	1154139.00
63	FusionSolutions63	729554.00
11	OmniTech11	108524.00
35	NextApps35	157820.00
47	SparkWorks47	6163067.00
14	NextLabs14	3740752.00
111	AlphaSystems111	771387.00
93	EcoSystems93	4270085.00
22	FusionTech22	2040563.00

6. LEFT OUTER JOIN

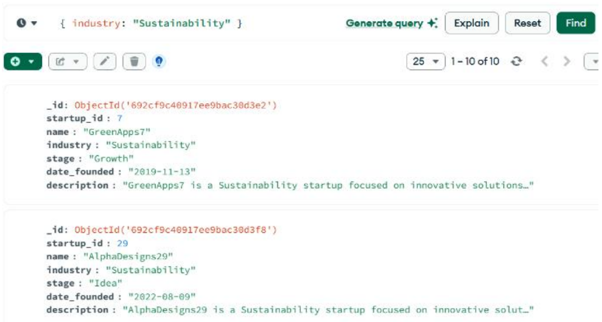
Query: Show all startups and any funding rounds they have (including startups with no funding)

```
SELECT
    s.startup_id,
    s.name AS startup_name,
    fr.funding_id,
fr.amount
FROM Startup s
LEFT JOIN FundingRound fr
ON s.startup_id = fr.startup_id
ORDER BY s.startup_id;
GROUP BY s.startup_id, s.name
ORDER BY total_funding DESC;
```

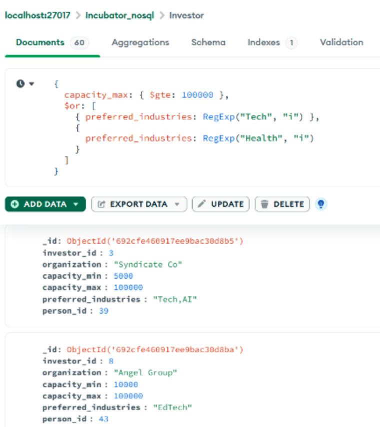
startup_id	startup_name	funding_id	amount
1	BlueSolutions1	141	46815.00
2	BlueApps2	75	29881.00
3	FusionDesigns3	88	177259.00
4	BrightApps4	NULL	NULL
5	EcoTech5	NULL	NULL
6	FusionWorks6	34	11396.00
6	FusionWorks6	69	1209768.00
6	FusionWorks6	101	9887.00
7	GreenApps7	3	23405.00
7	GreenApps7	127	748837.00
8	GreenWorks8	99	345169.00

<u>NoSQL Implementation:</u>

1. SIMPLE QUERY
Find all startups in the Sustainability industry

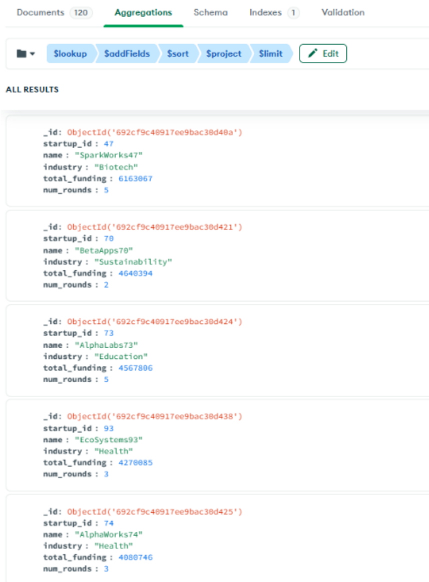


2. COMPLEX QUERY (multiple conditions)
Find investors who can invest at least $100,000 and prefer “Tech” or “Health” industries



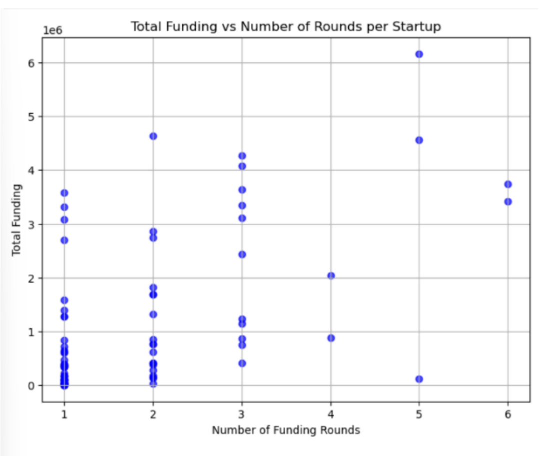
3. AGGREGATE QUERY
Top 10 startups by total funding received

```
[
{
$lookup: {
from: "FundingRound",
localField: "startup_id",
foreignField: "startup_id",
as: "funding"
}
},
{
$addFields: {
total_funding: { $sum: "$funding.amount" },
num_rounds: { $size: "$funding" }
}
},
{ $sort: { total_funding: -1 } },
{
$project: {
startup_id: 1,
name: 1,
industry: 1,
total_funding: 1,
num_rounds: 1
}
},
{ $limit: 10 }
]
```

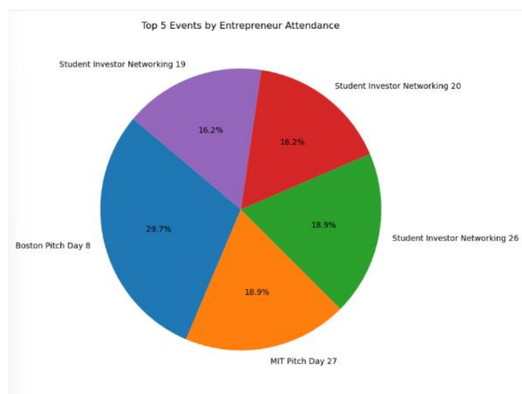


V. Database Access via Python

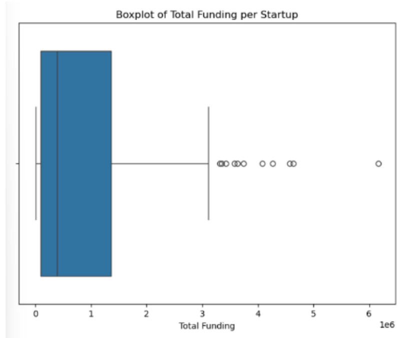
The database is accessed using Python and visualization of analyzed data is shown below. The connection of MySQL to Python is done using mysql.connector, followed by cursor.excecute to run and fetchall from query, followed by converting the list into a dataframe using pandas library and using matplotlib to plot the graphs for the analytics.



1. Scatter Plot: Total Funding vs Number of Rounds per Startup



2. Pie Chart: Event Participation (Top 5 Events by Attendees)



3. Boxplot: Total Funding per Startup

VI. Summary and Recommendation

The Boston Women Entrepreneurship Incubator Tracker designed in MySQL is a robust, industry-ready relational database tailored for incubators, accelerators, and university innovation hubs across Boston. The system centralizes information on women student entrepreneurs, their startups, mentors, investors, events, and funding trajectories, areas that were previously fragmented, inconsistent, and difficult to analyze. By unifying this data, the platform significantly streamlines the workflow for incubators and provides deep, actionable insights into the regional entrepreneurship ecosystem.

The analytics component further enhances the system's value by enabling incubators to understand which startups are progressing rapidly, which mentors are the most active or influential, which industries show strong founder participation, and which investors consistently support women-led ventures.

A key direction for improvement is the implementation of comprehensive data governance practices. Since multiple stakeholders (entrepreneurs, mentors, investors, incubator staff) contribute data, strong governance is essential to ensure quality, accuracy, and consistency across all entries. Features such as identity verification for mentors and investors, automated duplicate detection for startups, and standardized classification for industries and event types could greatly enhance reliability.

Although a partial Neo4j NoSQL prototype shows promising support for graph-style relationship queries such as mapping mentor networks or investor-founder relationship patterns, further exploration is needed to determine whether a full migration or hybrid architecture would meaningfully improve performance.

Overall, the Boston Women Entrepreneurship Incubator Tracker lays a strong foundation for transforming how incubators support women founders in Boston. With additional refinement, governance enhancements, and possibly hybrid database architecture, the system holds strong potential to be adopted across universities and accelerator programs, ultimately creating a more transparent, data-driven, and equitable innovation ecosystem.